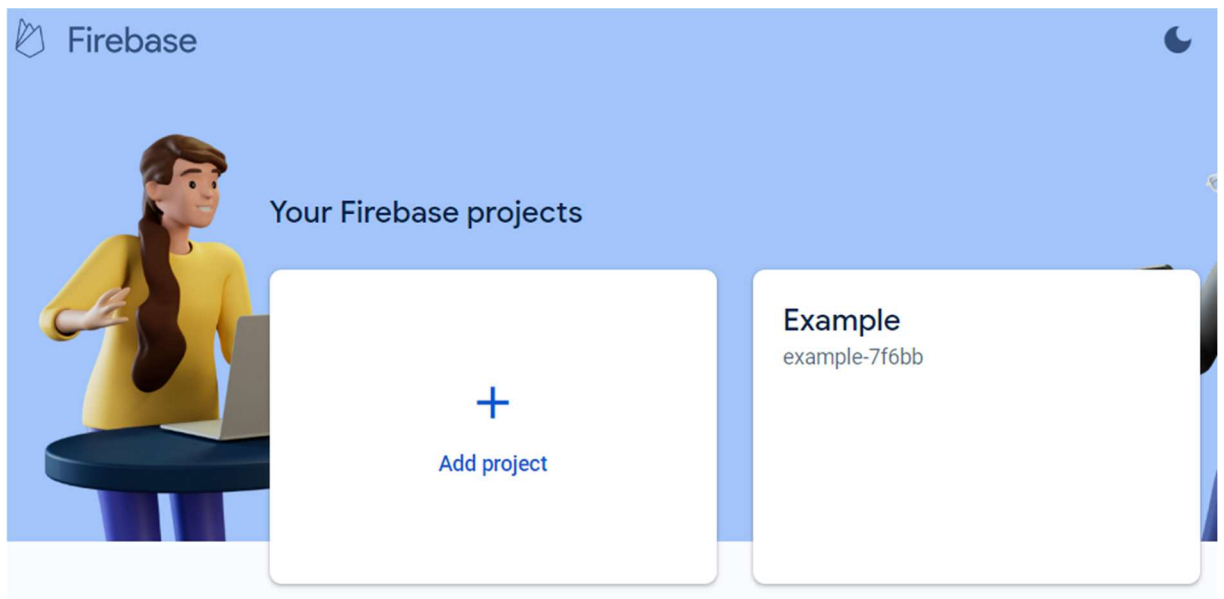


Screenshots liés à l'installation de Firebase avec un projet Flutter

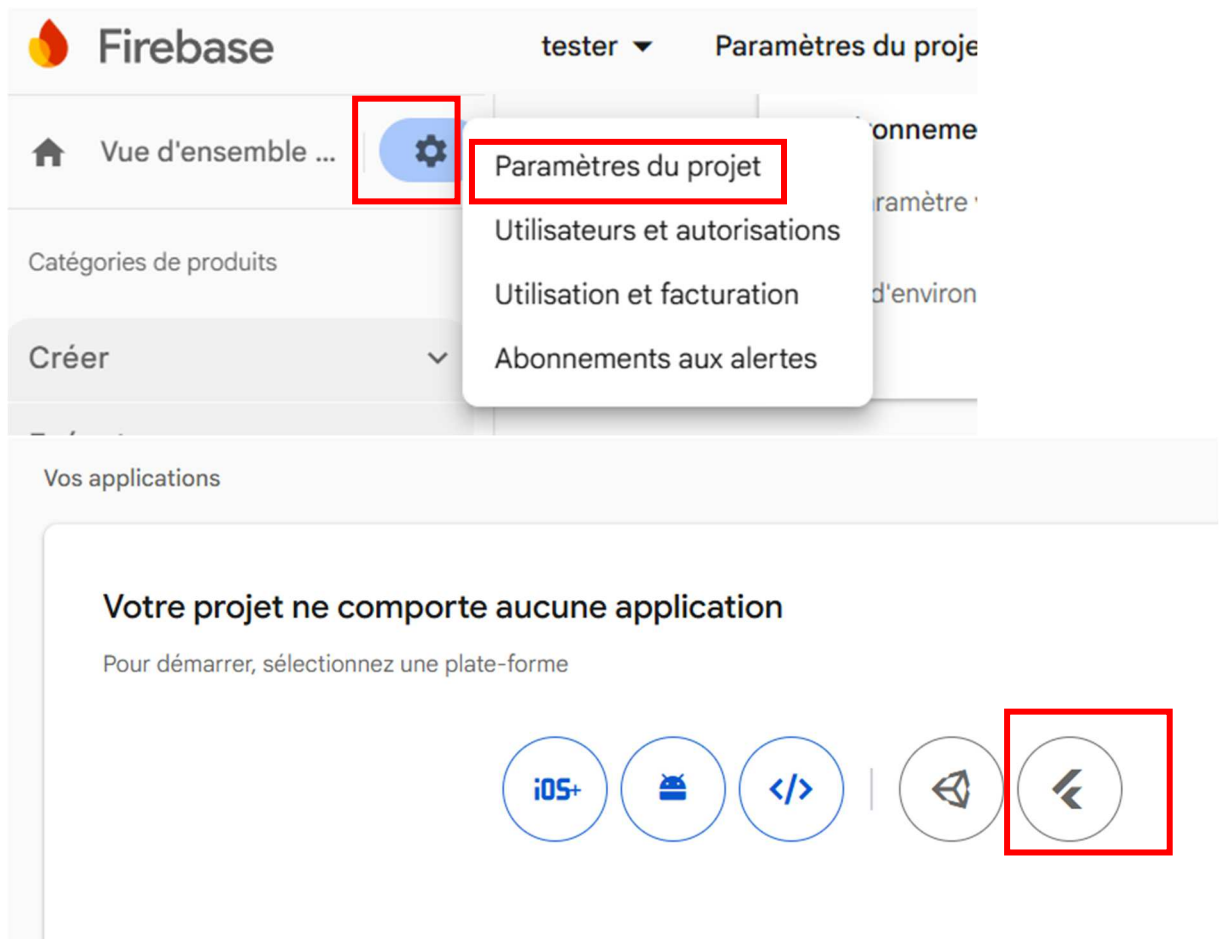
- 1- Création d'un compte Firebase (possibilité de se connecter avec Gmail) sur [www.firebase.google.com](https://firebase.google.com) :



- 2- Création d'un projet Firebase (depuis le bouton Go to console) :



- 3- Choisir l'option configuration pour Flutter :



4- Installez Firebase CLI depuis le lien de la figure suivante :

× Add Firebase to your Flutter app

1 Prepare your workspace

The easiest way to get you started is to use the FlutterFire CLI.

Before you continue, make sure to:

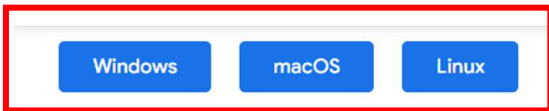
- Install the [Firebase CLI](#) and log in (run `firebase login`)
- Install the [Flutter SDK](#)
- Create a Flutter project (run `flutter create`)

Next

2 Install and run the FlutterFire CLI

3 Initialize Firebase and add plugins

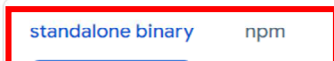
- a. Choisir votre OS ainsi que l'une des options d'installation :



Windows

You can install the Firebase CLI for Windows using one of the following options:

Option	Description	Recommended for...
standalone binary	Download the standalone binary for the CLI. Then, you can access the executable to open a shell where you can run the <code>firebase</code> command.	New developers Developers not using or unfamiliar with Node.js ↗
npm	Use npm (the Node Package Manager) to install the CLI and enable the globally available <code>firebase</code> command.	Developers using Node.js ↗



To download and run the binary for the Firebase CLI, follow these steps:

1. Download the [Firebase CLI binary for Windows](#) [↗](#).
2. Access the binary to open a shell where you can run the `firebase` command.
3. Continue to [log in and test the CLI](#).

- b. En choisissant l'OS Windows et l'option `npm`, il faut installer Node.js. Puis installer Firebase-tools depuis le terminal (CMD) de Windows en utilisant la commande suivante :

standalone binary

npm

To use `npm` (the Node Package Manager) to install the Firebase CLI, follow these steps:

1. Install [Node.js](#) [↗](#) using [nvm-windows](#) [↗](#) (the Node Version Manager). Installing Node.js automatically installs the `npm` command tools.

★ **Note:** The Firebase CLI requires **Node.js v18.0.0 or later**. Some Firebase features might require specific versions of Node.js, so check each Firebase product's getting started page for any specific Node.js requirements.

2. Install the Firebase CLI via `npm` by running the following command:

```
$ npm install -g firebase-tools
```

This command enables the globally available `firebase` command.

★ **Note:** If the `npm install -g firebase-tools` command fails, you might need to [change npm permissions](#) [↗](#).

3. Continue to [log in and test the CLI](#).

Deux options
possibles

Node.js® is an open-source, cross-platform JavaScript runtime environment.

[New security releases to be made available February 6th, 2024](#) ↗

Download Node.js®

20.11.0 LTS

Recommended For Most Users

21.6.1 Current

Latest Features

[Other Downloads](#) | [Changelog](#) | [API Docs](#) [Other Downloads](#) | [Changelog](#) | [API Docs](#)

For information about supported releases, see the [release schedule](#).

Redémarrez la machine et vérifiez que Node.js a bien été installé

```
PS C:\Users\PcTec> node --version  
v20.11.0  
PS C:\Users\PcTec>
```

```
Invite de commandes  X + v  
Microsoft Windows [version 10.0.22621.3007]  
(c) Microsoft Corporation. Tous droits réservés.  
C:\Users\PcTec> npm install -g firebase-tools  
added 640 packages in 1m  
  
65 packages are looking for funding  
  run 'npm fund' for details  
npm notice  
npm notice New minor version of npm available! 10.2.4 -> 10.4.0  
npm notice Changelog: https://github.com/npm/cli/releases/tag/v10.4.0  
npm notice Run npm install -g npm@10.4.0 to update!  
npm notice  
C:\Users\PcTec>
```

Acceptez la mise à jour proposée :

```
Invite de commandes X + v
Microsoft Windows [version 10.0.22621.3007]
(c) Microsoft Corporation. Tous droits réservés.

C:\Users\PcTec>npm install -g firebase-tools

added 640 packages in 1m

65 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New minor version of npm available! 10.2.4 -> 10.4.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v10.4.0
npm notice Run npm install -g npm@10.4.0 to update!
npm notice

C:\Users\PcTec>

C:\Users\PcTec>npm install -g npm@10.4.0
[ ] \ reify:npm: timing reify:createSparse Completed in 5ms
added 1 package in 8s
24 packages are looking for funding
  run `npm fund` for details

C:\Users\PcTec>
```

Vérifiez que Firebase a bien été installé (si nécessaire redémarrez la machine avant) :

```
C:\WINDOWS\system32\cmd. X + v

C:\Users\PcTec>firebase --version
13.1.0

C:\Users\PcTec>
```

Se connecter à Firebase :

```
C:\WINDOWS\system32\cmd. X + v

C:\Users\PcTec>firebase --version
13.1.0

C:\Users\PcTec>firebase login
i Firebase optionally collects CLI and Emulator Suite usage and error reporting information to help
  icy (https://policies.google.com/privacy) and is not used to identify you.

? Allow Firebase to collect CLI and Emulator Suite usage and error reporting information? (Y/n) y
```

Woohoo!

Firebase CLI Login Successful

You are logged in to the Firebase Command-Line interface. You can immediately close this window and continue using the CLI.

Testez la connexion en listant les projets Firebase créés sur la console du compte Firebase :

```
Waiting for authentication...
+ Success! Logged in as asma.mehiaoui.hamitou@gmail.com
C:\Users\PcTec>firebase projects:list
✓ Preparing the list of your Firebase projects
```

Project Display Name	Project ID	Project Number	Resource Location ID
Example	example-7f6bb	878217423393	[Not specified]
Exemple	exemple-b6169	773270430996	[Not specified]

```
2 project(s) total.
```

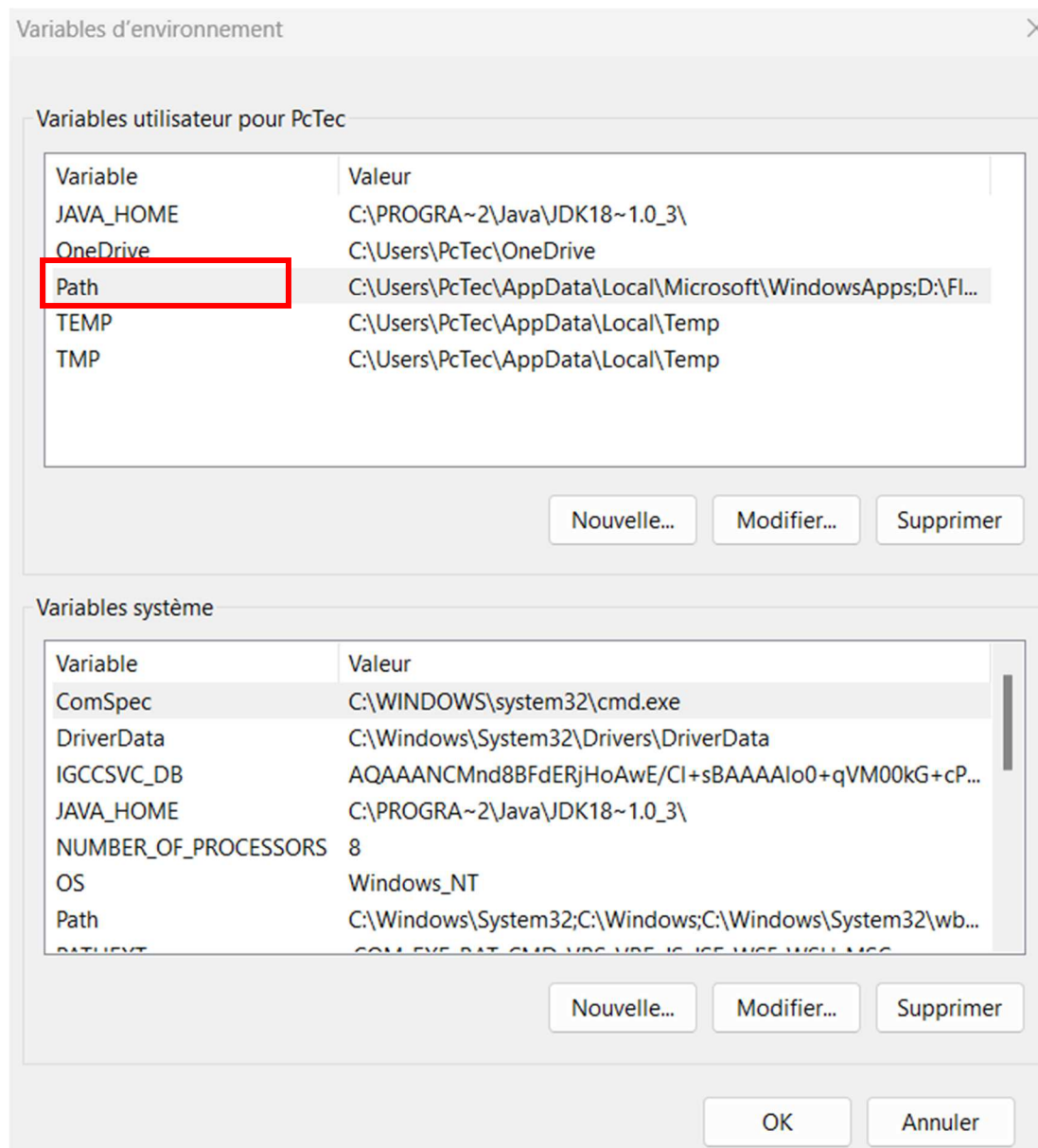
- 5- Installez FlutterFire_CLI (Cette commande on peut l'exécuter depuis le terminal windows ou le terminal vscode) :

```
C:\WINDOWS\system32\cmd. X + v

C:\Users\PcTec>dart pub global activate flutterfire_cli
+ ansi_styles 0.2.2 (2.1.1 available)
+ args 2.4.2
+ async 2.11.0
+ boolean_selector 2.1.1
+ characters 1.3.0
+ ci 0.1.0
+ cli_util 0.3.5 (0.4.1 available)
+ clock 1.1.1
+ collection 1.18.0
+ dart_console 1.2.0
+ deep_pick 0.10.0 (1.0.0 available)
+ ffi 2.1.0 (2.1.2 available)
+ file 6.1.4 (7.0.0 available)
+ flutterfire_cli 0.2.7
+ http 0.13.6 (1.2.0 available)
+ http_parser 4.0.2
+ interact 2.2.0
+ intl 0.18.1 (0.19.0 available)
+ json_annotation 4.8.1
+ matcher 0.12.16+1
+ meta 1.11.0
+ path 1.9.0
+ petitparser 6.0.1 (6.0.2 available)
+ platform 3.1.4
+ process 4.2.4 (5.0.2 available)
+ pub_semver 2.1.4
+ pub_updater 0.2.4 (0.4.0 available)
+ pubspec 2.3.0
+ quiver 3.2.1
+ source_span 1.10.0
+ stack_trace 1.11.1
+ stream_channel 2.1.2
+ string_scanner 1.2.0
+ term_glyph 1.2.1
+ test_api 0.7.0
+ tint 2.0.1
+ typed_data 1.3.2
+ uri 1.0.0
+ win32 5.1.1 (5.2.0 available)
+ xml 6.4.2 (6.5.0 available)
+ yaml 3.1.2
Building package executables... (7.1s)
Built flutterfire_cli:flutterfire.
Installed executable flutterfire.
Warning: Pub installs executables into C:\Users\PcTec\AppData\Local\Pub\Cache\bin, which is not on your path.
You can fix that by adding that directory to your system's Path environment variable.
A web search for "configure windows path" will show you how.
Activated flutterfire_cli 0.2.7.

C:\Users\PcTec>
```

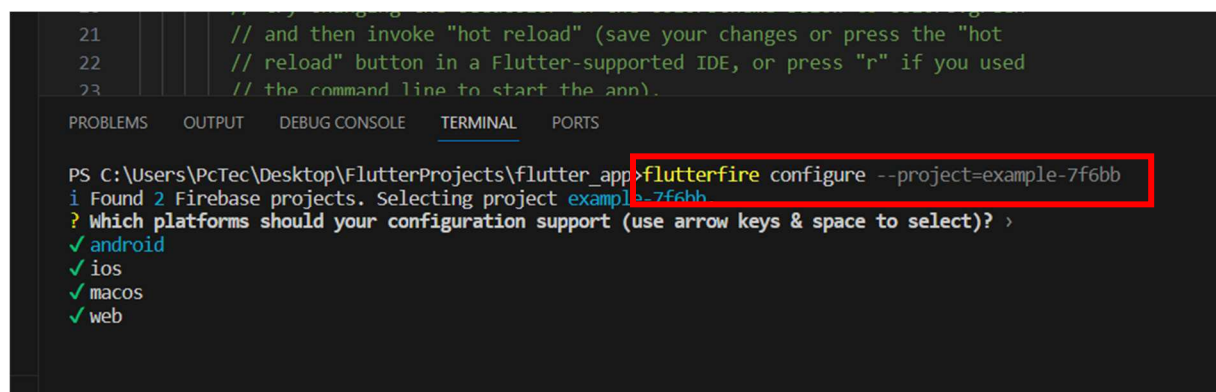
Ajoutez FlutterFire_CLI à la variable d'environnement de Windows :



Redémarrez la machine.

- 6- Exécutez FlutterFire pour générer le fichier de configuration : `flutterfire configure --project=example-7f6bb` il faut la récupérer depuis le site de firebase et il faut l'exécuter depuis le projet flutter.

Utilisez l'espace pour sélectionner ou désélectionner un choix.



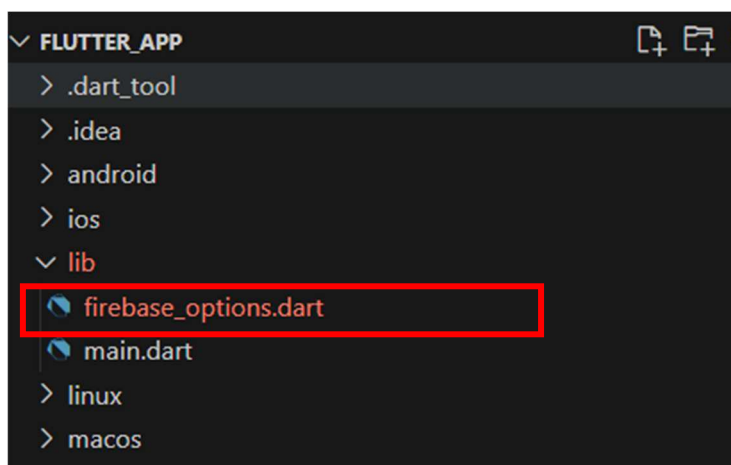
```
10 // This widget is the root of your application.
11 @override
12 Widget build(BuildContext context) {
13   // This method is required by FlutterFire to generate the configuration file.
14   // You can find more information about this method in the documentation:
15   // https://firebase.google.com/docs/flutter/setup
16   return MaterialApp(
17     title: 'Flutter Demo',
18     theme: ThemeData(
19       primarySwatch: Colors.blue,
20     ),
21     home: _HomePage(),
22   );
23 }
24
25 PS C:\Users\PcTec\Desktop\FlutterProjects\flutter_app>flutterfire configure --project=example-7f6bb
i Found 2 Firebase projects. Selecting project example-7f6bb.
✓ Which platforms should your configuration support (use arrow keys & space to select)? · ios, android
i Firebase android app com.example.flutter_app is not registered on Firebase project example-7f6bb.
i Registered a new Firebase android app on Firebase project example-7f6bb.
i Firebase ios app com.example.flutterApp is not registered on Firebase project example-7f6bb.
i Registered a new Firebase ios app on Firebase project example-7f6bb.

Firebase configuration file lib\firebase_options.dart generated successfully with the following Firebase apps:

Platform  Firebase App Id
android   1:878217423393:android:772de7753b581305d81160
ios       1:878217423393:ios:d00c90fefe5d7e76d81160

Learn more about using this file and next steps from the documentation:
> https://firebase.google.com/docs/flutter/setup
PS C:\Users\PcTec\Desktop\FlutterProjects\flutter_app>
```

Cette dernière commande va générer le fichier dans le projet Flutter :



7- Installez firebase_core plugin pour résoudre les problèmes du fichier généré :

```
lib > firebase_options.dart > ...
1 // File generated by flutterfire CLI.
2 // ignore_for_file: lines_longer_than_80_chars, avoid_classes_with_only_static_members
3 import 'package:firebase_core/firebase_core.dart' show FirebaseOptions;
4 import 'package:flutter/foundation.dart'
5   show defaultTargetPlatform, kIsWeb, TargetPlatform;
6
7 /// Default [FirebaseOptions] for use with your Firebase apps.
8 ///
```

```
PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\PcTec\Desktop\FlutterProjects\flutter_app> flutter pub add firebase_core
```

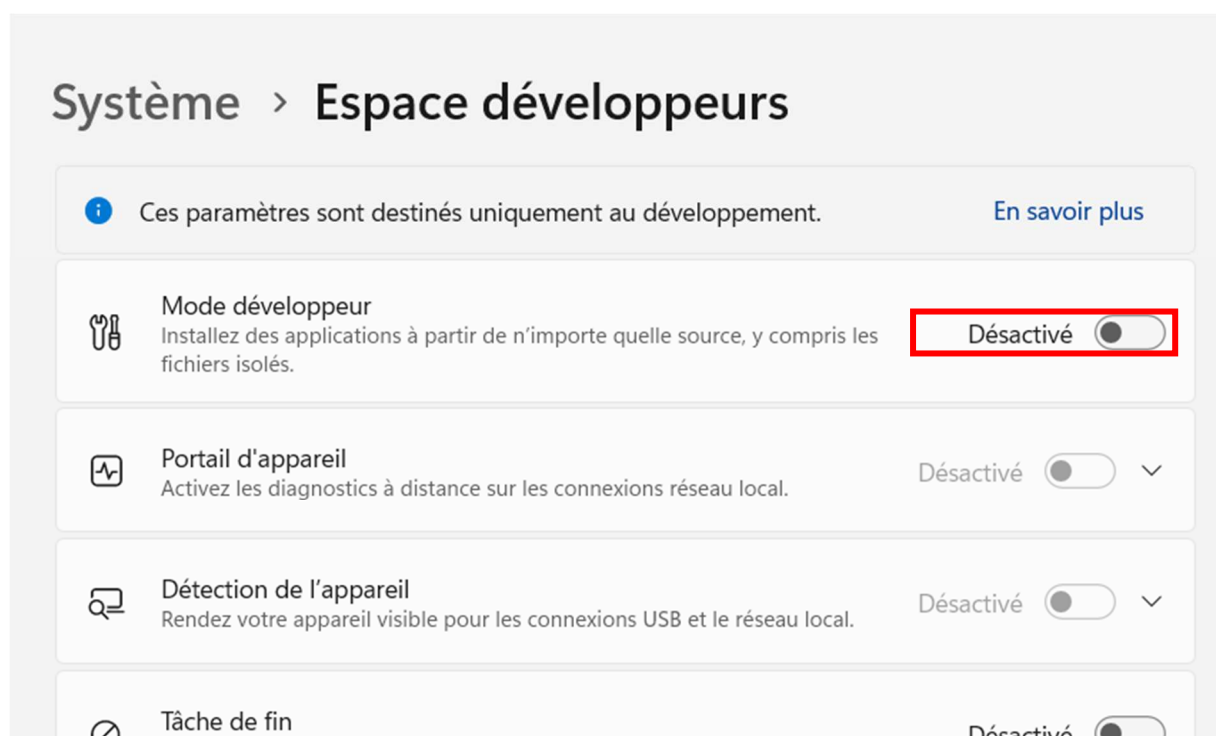
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\PcTec\Desktop\FlutterProjects\flutter_app> flutter pub add firebase_core
Resolving dependencies...
  collection 1.17.2 (1.18.0 available)
+ firebase_core 2.24.2 (2.25.3 available)
+ firebase_core_platform_interface 5.0.0
+ firebase_core_web 2.10.0 (2.11.3 available)
  flutter_lints 2.0.3 (3.0.1 available)
+ flutter_web_plugins 0.0.0 from sdk flutter
+ js 0.6.7 (0.7.0 available)
  lints 2.1.1 (3.0.0 available)
  matcher 0.12.16 (0.12.16+1 available)
  material_color_utilities 0.5.0 (0.8.0 available)
  meta 1.9.1 (1.11.0 available)
  path 1.8.3 (1.9.0 available)
+ plugin_platform_interface 2.1.8
  stack_trace 1.11.0 (1.11.1 available)
  stream_channel 2.1.1 (2.1.2 available)
  test_api 0.6.0 (0.7.0 available)
  web 0.1.4-beta (0.4.2 available)
Changed 6 dependencies!
Building with plugins requires symlink support.

Please enable Developer Mode in your system settings. Run
start ms-settings:developers
to open settings.
PS C:\Users\PcTec\Desktop\FlutterProjects\flutter_app> 
```

Activez le mode développeur sur le système si nécessaire :

```
PS C:\Users\PcTec\Desktop\FlutterProjects\flutter_app> start ms-settings:developers
```



```
dependencies:
  flutter:
    sdk: flutter
  cupertino_icons: ^1.0.2
  firebase_core: ^2.24.2
```

- 8- Importez le package `firebase_core` ainsi que le fichier de configuration généré depuis le fichier `main` de Flutter :

```
! pubspec.yaml ● main.dart 2 X
lib > main.dart > MyApp > build
1 import 'package:flutter/material.dart';
2 import 'package:firebase_core/firebase_core.dart';
3 import 'firebase_options.dart';
4
```

- 9- Initialisation de Firebase depuis le projet Flutter avant de lancer l'application (dans les grands projets cette initialisation n'est pas obligée d'être dans le `main`) :

```
Run | Debug | Profile
void main() async{
  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );

  runApp(const MyApp());
}
```

Pour l'autre option d'installation (c'est-à-dire en passant par un outil externe au lieu de `npm`), la seule différence est qu'il faut télécharger un exécutable et exécuter les commandes depuis cet exécutable.

////////////////////////////////////

Sur Mac

////////////////////////////////////

- 1) Installer Firebase CLI depuis le lien de la figure suivante :
 - a. Exécuter la commande tout en un depuis un terminal MacOS :


```
curl -sL https://firebase.tools | bash
```

× Add Firebase to your Flutter app

1 Prepare your workspace

The easiest way to get you started is to use the FlutterFire CLI.

Before you continue, make sure to:

- Install the [Firebase CLI](#) and log in (run `firebase login`)
- Install the [Flutter SDK](#)
- Create a Flutter project (run `flutter create`)

Next

2 Install and run the FlutterFire CLI

3 Initialize Firebase and add plugins

```
asma — -bash — 80x24
Last login: Tue Jan 30 20:49:50 on ttys018

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
MacBook-Pro-de-Asma:~ asma$ curl -sL https://firebase.tools | bash
Password:
-- Checking for existing firebase-tools on PATH...
-- Checking your machine type...
-- Downloading binary from https://firebase.tools/bin/macos/latest
##### 100.0%#
##### 100.0%
-- Setting permissions on binary...
-- Checking your PATH variable...
-- firebase-tools@13.1.0 is now installed
-- All Done!
MacBook-Pro-de-Asma:~ asma$ curl -sL https://firebase.tools | bash
```

- b. Executer la commande pour se connecter à Firebase:

```
firebase login
```

```
MacBook-Pro-de-Asma:~ asma$ firebase login  
i Firebase optionally collects CLI and Emulator Suite usage and error reporting  
information to help improve our products. Data is collected in accordance with  
Google's privacy policy (https://policies.google.com/privacy) and is not used to  
identify you.  
  
? Allow Firebase to collect CLI and Emulator Suite usage and error reporting  
information? (Y/n)
```

- c. Tester la connexion en affichant la liste des projets Firebase en exécutant la commande :

```
firebase projects:list
```

- 2) Installer FlutterFire_CLI en exécutant la commande :
`dart pub global activate flutterfire_cli`

2

Install and run the FlutterFire CLI

From any directory, run this command:

```
$ dart pub global activate flutterfire_cli
```

Then, at the root of your Flutter project directory, run this command:

```
$ flutterfire configure --project=firstproject-665ec
```

This automatically registers your per-platform apps with Firebase and adds a `lib/firebase_options.dart` configuration file to your Flutter project.

Previous

Next

```
MacBook-Pro-de-Asma:first_app asma$ dart pub global activate flutterfire_cli
```

The Dart tool uses Google Analytics to report feature usage statistics and to send basic crash reports. This data is used to help improve the Dart platform and tools over time.

To disable reporting of analytics, run:

```
dart --disable-analytics
```

```
+ ansi_styles 0.3.2+1s... (1.9s)
+ args 2.4.2
+ async 2.11.0
+ boolean_selector 2.1.1
+ characters 1.3.0
+ ci 0.1.0
+ cli_util 0.3.5 (0.4.1 available)
+ clock 1.1.1
+ collection 1.18.0
+ dart_console 1.2.0
+ deep_pick 0.10.0 (1.0.0 available)
+ ffi 2.1.0 (2.1.2 available)
+ file 6.1.4 (7.0.0 available)
+ flutterfire_cli 0.2.7
+ http 0.13.6 (1.2.0 available)
+ http_parser 4.0.2
+ interact 2.2.0
+ intl 0.18.1 (0.19.0 available)
+ json_annotation 4.8.1
+ matcher 0.12.16+1
+ meta 1.11.0
+ path 1.9.0
+ petitparser 6.0.1 (6.0.2 available)
+ platform 3.1.4
+ process 4.2.4 (5.0.2 available)
+ pub_semver 2.1.4
+ pub_updater 0.2.4 (0.4.0 available)
+ pubspec 2.3.0
+ quiver 3.2.1
+ source_span 1.10.0
+ stack_trace 1.11.1
+ stream_channel 2.1.2
+ string_scanner 1.2.0
+ term_glyph 1.2.1
+ test_api 0.7.0
+ tint 2.0.1
+ typed_data 1.3.2
+ uri 1.0.0
+ win32 5.1.1 (5.2.0 available)
+ xml 6.4.2 (6.5.0 available)
```

```
+ test_api 0.7.0
+ tint 2.0.1
+ typed_data 1.3.2
+ uri 1.0.0
+ win32 5.1.1 (5.2.0 available)
+ xml 6.4.2 (6.5.0 available)
+ yaml 3.1.2
```

```
Building package executables... (5.9s)
```

```
Built flutterfire_cli:flutterfire.
```

```
Installed executable flutterfire.
```

```
Warning: Pub installs executables into $HOME/.pub-cache/bin, which is not on your path.
You can fix that by adding this to your shell's config file (.bashrc, .bash_profile, etc.):
```

```
export PATH="$PATH":"$HOME/.pub-cache/bin"
```

```
Activated flutterfire_cli 0.2.7.
```

Modifier la variable d'environnement (PATH) en executants la commande suivante:

```

Activated flutterfire_cli 0.2.7.
MacBook-Pro-de-Asma:first_app asma$ export PATH="$PATH":"$HOME/.pub-cache/bin"
MacBook-Pro-de-Asma:first_app asma$

```

- 3) Le reste des étapes est la même chose que pour Windows (c'est la commande qui va s'exécuter pour chaque projet et chaque configuration différente) :

a. flutterfire configure --project=firstproject-665ec

```

i Firebase android app com.example.first_app is not registered on Firebase project firstproject-665ec.
i Registered a new Firebase android app on Firebase project firstproject-665ec.
i Firebase ios app com.example.firstApp is not registered on Firebase project firstproject-665ec.
i Registered a new Firebase ios app on Firebase project firstproject-665ec.
i Firebase macos app com.example.firstApp.RunnerTests is not registered on Firebase project firstproject-665ec.
i Registered a new Firebase macos app on Firebase project firstproject-665ec.
i Firebase web app first_app (web) is not registered on Firebase project firstproject-665ec.
i Registered a new Firebase web app on Firebase project firstproject-665ec.
? The files android/build.gradle & android/app/build.gradle will be updated to apply Firebase c
? The files android/build.gradle & android/app/build.gradle will be updated to apply Firebase c
✓ The files android/build.gradle & android/app/build.gradle will be updated to apply Firebase c
uild plugins. Do you want to continue? · yes

Firebase configuration file lib/firebase_options.dart generated successfully with the following

Platform  Firebase App Id
web       1:883718173739:web:75eb39f9d8dc6f93bbe66e
android   1:883718173739:android:f41132b57263329abb66e
ios       1:883718173739:ios:eac231b253e8ffd0bbe66e
macos     1:883718173739:ios:d5b5c4382152fafabb66e

Learn more about using this file and next steps from the documentation:
> https://firebase.google.com/docs/flutter/setup
MacBook-Pro-de-Asma:first_app asma$

```

```

ios
├── Flutter
├── Runner
├── Runner.xcodeproj
├── Runner.xcworkspace
├── RunnerTests
├── .gitignore
└── {} firebase_app_id_file.json

```

```

android
├── app
│   ├── src
│   └── {} google-services.json

```

- Installation du package `firebase_core`.
- Importation du package `firebase_core` & du fichier de configuration.
- Initialisation dans le `main.dart`.

Il se peut que votre anti virus (parefeu) bloque l'installation.